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Topic: Analytics

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~~Reasons~~ Reasoning

Section: 2

LR Assignment

Q1/

Five persons - Sandeep, Krish, Rostam, Harsh & Vinay, have some amount of money with them. Total amount of money they have is ₹450. Each of one of them has atleast ₹30.

Total amount of money that Sandeep has is twice the square of a natural number is over ₹100.

Harsh and Vinay have the amount of money in the ratio of 3:5.

Rostam has ₹4 more than twice the square of a natural number.

Total amount of money with Harsh and Vinay are equal to the amount of money with Sandeep.

Krish has an odd amount of money, less than Rostam.

1) Who among the following has highest amount of money.

A) Sandeep

B) Rostam

C) Harsh

D) Krish

E) Can't be determined.

2) What is the difference between the amount which Rostam and Krish have?

- A) ₹119
- B) ₹98
- C) ₹116
- D) ₹108
- E) Can't be determined.

3) Which among the following has least amount of money?

- A) Krish
- B) Harsh
- C) Vinay
- D) Either Harsh or Krish
- E) Either Krish or Rostam

4) What is the amount of money that Sandeep has?

- A) ₹200
- B) ₹160
- C) ₹128
- D) Either ₹200 or ₹128
- E) Either ₹80 or ₹160

5) What is the sum of money that ~~Harsh~~ Harsh and Sandeep have?

- A) ₹176
- B) ₹240
- C) ₹168
- D) ₹282
- E) Either ₹168 or ₹208

Solution

For sandeep

$$2n^2 > 100$$

$$n^2 > 50$$

$$n > 7$$

$\rightarrow$ 8, 9, 10 possibility

$$\text{Harsh} + \text{Vinay} = \text{sandeep}$$
$$\frac{\text{Harsh}}{\text{Vinay}} = \frac{3}{5}$$

Case-1 sandeep has $2 \times 8^2 = ₹128$.

then money with ~~Rustam & Krish~~ Rustam & Krish is

$$450 - 128 < 128 = ₹194$$

$\downarrow$
Harsh
+
Vinay

This is possible.

Case-2 sandeep has $2 \times 9^2 = ₹162$.

If sandeep has ₹162 then Harsh & Vinay have ₹162 then
the money with Rustam & Krish = ₹126.

Case-3 sandeep has $2 \times 10^2 = ₹200$.

If sandeep has ₹200 then Harsh & Vinay have ₹200
then the money with Rustam & Krish is ₹50.

$\downarrow$
which is not possible $\rightarrow$ as each has at least ₹30.

Now Harsh should be $\frac{3}{5}$

$$\frac{H}{V} = \frac{3}{5}$$

If $H + V = 128$

$$\frac{3V}{5} + V = 128$$

$$\frac{8V}{5} = 128$$

$$V = 80$$

$$H = 48$$

$$8x = 128$$

$$x = 16$$

$$x = 16$$

$$3x = 48$$

Harsh

$$5x = 80$$

Vinay

If $H + V = 162$

$$\frac{48V}{5} = 162$$

$$V = \frac{10}{4}$$

$$V = 405$$

$$8x = 162$$

$$x = 20.25$$

Harsh

not a natural number

Therefore possibility rejected

currently

Sandeep $\rightarrow$ ₹128

Vinay $\rightarrow$ ₹80

Harsh $\rightarrow$ ₹48

Total money with Rastam and Krish is ₹194.

Krish has odd amount of money less than Rastam.

Case-A

$$m=3$$

Money with Rastam

$$\text{Le } 3 \times 3^2 + 4 = ₹31$$

Money with Krish $\rightarrow ₹153$

$$\text{Money with Rastam} \rightarrow 4 + 3m^2$$

$$4 + 3m^2 \geq 30$$

$$3m^2 \geq 26$$

$$m^2 \geq$$

$$m \geq 3$$

not possible
as Krish
has less money than
Rastam

Case-B

$$m=4, m=5, m=6$$

Not possible for the same above reason

Case-C

$$m=7$$

$$\text{Rastam has} \rightarrow 3 \times 7^2 + 4 = ₹151$$

Krish has ₹43.

Rastam $\rightarrow ₹151$

Sandeep $\rightarrow ₹128$

Vinay $\rightarrow ₹80$

Rash $\rightarrow ₹48$

Krish $\rightarrow ₹43$

Answers

1 $\rightarrow$ B

2 $\rightarrow$ D

3 $\rightarrow$ A

4 $\rightarrow$ C

5 $\rightarrow$ A

Q2/ Seven persons - Subh, Isha, Kanya, Ritu, Manav, Drona and Uttam are divided into 3 teams - A, B, C. There are at least 2 persons and only male persons included in each team. Among these persons two of them belong to Delhi, two belong to Noida and 3 belong to Agra.

Kanya belongs to Noida and she is not in the same team as pair of sisters Subh and Drona.

Isha is a male, who belongs to Delhi, and is in the Team - A with only Uttam, who belongs to Agra.

Ritu is a male and belongs to Noida.

Two persons belonging to the same city were not of the same team.

Sibh does not belong to Delhi, but was part of Team - B

1) Who among the following is a male in Team C?

- A) Ritu
- B) Manav
- C) Kangs
- D) Druv
- E) Can't be determined

2) Which of the following combinations show the males among the persons?

- A) Isha and Ritu
- B) Druv, Ritu and Isha
- C) Manav, Ritu and Kangs
- D) Manav, Ritu and Isha
- E) Isha, Manav and Druv

3) Manav belongs to which of the following cities?

- A) Agra
- B) Delhi
- C) Noida
- D) Either Agra or Delhi
- E) Either Delhi or Noida

4) Which of the following combinations is/are correct?

- A) Manav - Female - Agra
- B) Sibh - Male - Agra
- C) Ritu - Male - Delhi
- D) All are correct
- E) None

5) How many females belong to Agra?

- A) One
- B) Two
- C) Three
- D) Either one or Two
- E) Either Two or Three

Solution Step-1

Team	Person	City
A	Ishat	Delhi
	Uttam	Agra
B	Subh	

Subh does not belong to Delhi

Step-2

Team	Person	City
A	Ishat	Delhi
	Uttam	Agra
B	Subh	
	Drna	
C	Kanya	Wade

Step-3

Team	Person	City
A	Isha +	Delhi
	Uttam -	Agra
B	Subh -	Moida
	Pransh - Ritu +	
C	Kanya -	Moida

Step-4

Team	Person	City
A	Isha (+)	Delhi
	Uttam + -	Agra
B	Subh -	Agra
	Pransh -	Delhi
	Ritu +	Moida
C	Kanya -	Moida
	Pransh +	Agra

Answer

1 → B

2 → D

3 → A

4 → E

5 → B